

INT 21: DOS functions

DOS 1.x

AH=00: Terminate Program CS->PSP	(=INT 20) restore INT 22,23,24 vectors from old values stored in PSP. does not close open files.
AH=01: Read Keyboard with Echo - input may be redirected	AL=InputChar =0 for extended keys (function keys etc) next call reads the scancode <ctrl-break>: INT 23
AH=02: Display Character - output may be redirected DL=OutputChar =7 (Beep) <ctrl-break>: INT 23	
AH=03: Auxiliary Input wait for input from standard device (COM1) - input may be redirected to other serial port (via MODE)	AL=InputChar <ctrl-break>: INT 23
AH=04: Auxiliary Output to standard device (COM1) - may be redirected DL=OutputChar <ctrl-break>: INT 23	
AH=05: Print Character DL=OutputChar - to standard printer (LPT1) <ctrl-break>: INT 23	see INT 17
AH=06: Direct Console I/O DL= FF: read from standard input (keyboard) - no wait - no echo - no check for <^break> < FF: write DL to standard output (screen)	read - character ready: Zf=0, AL=InputChar (=0: extended key - read again) read - character not ready: Zf=1, AL undefined
AH=07: Direct Console Input read from standard input (keyboard) - wait for key - no echo - no check for <^break>	AL=InputChar (=0: extended key - read again)
AH=08: Read Keyboard without Echo read from standard input - wait for key - no echo - check for <^break> (INT 23)	AL=InputChar (=0: extended key - read again)
AH=09: Display String DS:DX -> String\$ each char is sent as with function 02: Display character	
AH=0A: Buffered Keyboard Input read string from standard input - wait for input - echo - terminated by <enter> (used by COMMAND.COM at prompt) DS:DX -> buffer: byte[DX] : buffer length (1..255)	byte[DX+1]=bytes read (excl. terminal CR) [DX+2]=first byte terminal byte = 0D <Enter> if buffer is full: further input will be ignored and beeps emitted
AH=0B: check keyboard status	AL=00: buffer empty =FF: at least one character in buffer <ctrl-break>: INT 23
AH=0C: Flush Buffer, Read Keyboard empty standard input buffer, then: AL=01: INT 21,1: Read Keyboard with Echo =06: INT 21,6: Direct Console I/O =07: INT 21,7: Direct Console Input =08: INT 21,8: Read Keyboard without Echo =0A: INT 21,A: Buffered Keyboard Input	AL=InputChar

AH=0D: Reset disk drive ! files with changed size must be closed before the call ! - Flush all file buffers.	
AH=0E: select disk drive selected drive will be new default DL=driveID: 0=A, 1=B,,,	AL=total no. of logical drives
AH=0F: Open File DS:DX->FCB to be opened	AL=00: file opened - FCB filled =FF: error
AH=10: Close File DS:DX->open FCB	AL=00: file closed =FF: error
AH=11: Search first file match to filespec (wildcards ok) in FCB. Searches only current directory - for path-search, use INT 21,4E. Before this call: use INT 21,1A: Set DTA. DS:DX->FCB (not open)	AL=00: match found DTA filled with unopened FCB - opening the file destroys info for 'Search next' =FF: not found
AH=12: Search next file match Before this call: use previous DS:DX->FCB (not open)	AL=00: match found DTA filled with unopened FCB - opening the file destroys info for 'Search next' =FF: not found
AH=13: Delete file All files matching filespec in FCB are deleted. DS:DX->FCB (not open)	AL=00: file(s) deleted =FF: error
AH=14: Read File Sequential DS:DX->open FCB with - filename - record size - record counter	AL=00: file record read into memory at DTA FCB updated to next record =01: EOF (no data) =02: DTA too small =03: EOF (part of record read)
AH=15: Write File Sequential DS:DX->open FCB with - filename - record size - record counter	AL=00: file record written from DTA to buffer - whole sectors are written to disk FCB updated to next record =01: Disk full =02: DTA too small
AH=16: Create File DS:DX->FCB (not open) with filename	AL=00: File created with length 0 =FF: error
AH=17: Rename File with FCB DS:DX->FCB with new filespec	AL=00: File(s) renamed =FF: error
AH=19: Get Default Drive	AL=0,1,...: A,B,...
AH=1A: Set DTA-address (80h in PSP if not set) DS:DX->DTA-address	
AH=1B: Get FAT info for current (default) drive	AL=sectors pr cluster CX=sector size DX=number of clusters DS:BX->media descriptor byte
AH=1C: Get FAT info for specified drive DL=Drive number	AL=sectors pr cluster CX=sector size DX=number of clusters DS:BX->media descriptor byte
AH=21: Read Random Record DS:DX->open FCB	AL=00: Record read to DTA =01: EOF (no data) =02: DTA too small =03: EOF (part of record read)
AH=22: Write Random Record DS:DX->open FCB	AL=00: Record written from DTA =01: Disk full =02: DTA too small

AH=23: Get File Size no. of records DS:DX->FCB with - filespec - RecordSize	AL=00: no. of records returned in FCB =FF: error
AH=24: Set Random Record Number using CurBlockNo & CurRecNo DS:DX->open FCB	
AH=25: set interrupt vector AL=int no. DS:DX=new vector	
AH=26: Create new PSP with copy of calling .COM programs PSP (CS must point to this) DX=segment addr of new PSP	
AH=27: Random Block Read DS:DX->open FCB CX=number of records to be read	AL=00: Records read to DTA =01: EOF (no data) =02: DTA too small =03: EOF (part of record read) CX=number of records read
AH=28: Random Block Write DS:DX->open FCB CX=number of records to be written	AL=00: Records written from DTA =01: EOF (no data) =02: DTA too small CX=number of records written
AH=29: Parse Filename AL=Parsing control bits: see MS-DOS PR p.254 DS:SI->Command Line ("d:filename.ext'0'") wildcards ok, or several filenames separated by spaces ES:DI->FCB to be filled (not opened)	AL=00: no wildcards used =01: wildcards used =FF: drive not accepted DS:SI->1.char after filename ES:DI->1.byte of filled FCB
AH=2A: Get Date	AL=weekday (0=sunday) CX=year (1980-2099) DH=month (1-12) DL=day (1-31)
AH=2B: Set Date CX=year (1980-2099) DH=month (1-12) DL=day (1-31)	AL=00: no error AL=FF: error
AH=2C: Get Time	CH=hours (0-23) CL=minutes (0-59) DH=seconds (0-59) DL=sec/100 (0-99)
AH=2D: Set Time CH=hours (0-23) CL=minutes (0-59) DH=seconds (0-59) DL=sec/100 (0-99)	AL=00: no error AL=FF: error
AH=2E: Set Verify On/Off AL=0,1: set verify off,on	

DOS 2.x

AH=2F: Get DTA-address	ES:BX->DTA
AH=30: get DOS version number	AH=minor number AL=major number
AH=31: Keep Program AL=return code DX=size in paragraphs of code to remain in memory	
AH=33: Get/Set Ctrl-C check flag AL=00: Get AL=01: Set: DL=00,01: Set off, Set on	Get: DL=00,01: flag is off,on
AH=35: get interrupt vector AL=int no.	ES:BX=vector

AH=36: Get Disk Free Space DL=drive no. (0=default, 1=A,...)	AX=no. of sectors pr cluster =FFFF: error BX=no. of free clusters CX=no.of bytes pr sector DX=no. of clusters
AH=38: Set Country Code DX=FFFF IF CountryCode < 254: AL=CountryCode (45=DK) ELSE: AL=FF BX=CountryCode	
AH=38: Get Country Info DX<>FFFF: DS:DX->CountryInfo structure IF CountryCode < 254: AL=CountryCode (45=DK 0=current country) ELSE: AL=FF BX=CountryCode	CF=1: error AX=error code CF=0: AL=CountryCode (low 8 bits) (45=DK) BX=CountryCode DS:DX->filled CountryInfo structure
AH=39: Create Directory DS:DX->ASCIIZ DirSpec	CF=1: AX=error code
AH=3A: Remove Directory DS:DX->ASCIIZ DirSpec	CF=1: AX=error code
AH=3B: change current directory DS:DX->path0	CF=1: AX=error code
AH=3C: Create File with Handle (compare INT 21,5B) DS:DX->path0 CX=file attrib	IF file exists: no error CF=1: AX=error code CF=0: AX=Handle file opened
AH=3D: Open File with Handle AL=access code: 0=read 1=write 2=read/write DS:DX->FileSpec0	CF=1: AX=error code CF=0: AX=Handle file opened
AH=3E: close handle BX=handle	CF=1: AX=error code
AH=3F: read from file or device BX=handle (0=standard input: keybd) CX=byte count DS:DX->buffer	CF=1: AX=error code CF=0: AX=bytes read
AH=40: write to file or device BX=handle (1=standard output: scrn) CX=byte count DS:DX->buffer	CF=1: AX=error code CF=0: AX=bytes written
AH=41: Delete File DS:DX->FileSpec0	CF=1: AX=error code
AH=42: Move File Pointer AL=move method =00: absolute from file start (+) =01: relative from current position (+-) =02: absolute from file end (+-) BX=handle CX,DX=move distance (bytes)	CF=1: AX=error code CF=0: DX,AX=new pointer
AX=4300: Get File Attributes DS:DX-> full ASCIIZ filename	CF=1: AX=error code CF=0: CX=attributes
AX=4301: Set File Attributes CX=attributes DS:DX-> full ASCIIZ filename	

AH=44: I/O Device Control AL=00: Get device info =01: Set device info =02: Read from character device =03: Write to character device =04: Read from block device etc...	CF=1: AX=error code CF=0: DX=device info (for AL=00)
AH=45: Duplicate File Handle BX=handle	CF=1: AX=error code CF=0: AX=new handle
AH=46: Force Duplicate File Handle BX=handle CX=new handle	CF=1: AX=error code
AH=47: Get Current Directory DL=drive no. DS:SI->64 byte area	CF=1: AX=error code CF=0: DS:SI->filled 64 byte area
AH=48: Allocate Memory BX=requested paragraphs	CF=1: AX=error code CF=0: AX:00->allocated memory BX=allocated paragraphs
AH=49: Free Allocated Memory ES=segment of block to free	CF=1: AX=error code
AH=4A: Set Memory Block Size ES=segment of block to set (change) BX=requested paragraphs	CF=1: AX=error code CF=0: BX=allocated paragraphs
AH=4B: Load (and Execute) Program AL=00: Load and execute =03: Load overlay DS:DX->ASCIIIZ filespec ES:BX->parameter block	CF=1: AX=error code
AH=4C: terminate program AL='errornr' (used by ERRORLEVEL)	
AH=4D: Get Return Value from INT 21,31 or INT 21,4C - only once!	AH=return value AL=00: normal termination =01: ctrl-break termination =02: critical error =03: termination by INT 21,31
AH=4E: Find First File CX=attrib DS:DX->ASCIIIZ filespec	CF=1: AX=error code CF=0: DTA filled with Fileinfo struc.
AH=4F: Find Next File DTA contains Fileinfo	CF=1: AX=error code CF=0: DTA Fileinfo updated
AH=52: get list of lists (undoc) -PCM.10.24.466:CDS structure -GOOKIN: Enhanced...p325	ES:BX point to system info +16h point to first CDS +21h number of CDS structures set by LASTDRIVE - default: 5 (E)
AH=54: Get Verify State	AL=00: Verify is off =01: Verify is on
AH=56: Rename File - may be used to 'move' a file DS:DX->ASCIIIZ filespec ES:DI->ASCIIIZ new filespec	CF=1: AX=error code
AX=5700: Get File Date and Time BX=Handle	CF=1: AX=error code CF=0: CX=time (bits:) 15-11: hour (0-23) 10- 5: minute (0-59) 4- 0: 2sec (0-29) DX=date (bits:) 15- 9: year (since 1980) 8- 5: month (1-12) 4- 0: day (1-31)
AX=5701: Set File Date and Time BX=Handle CX=time, DX=date (as above)	CF=1: AX=error code

DOS 3.x

AH=59: Get Extended Error Info	AX=extended error code >PC'ere UL s.185 BH=error type BL=suggested action CH=location of error
AH=5A: Create 'Temporary' File CX=attrib DS:DX->ASCIIIZ dirs spec (with terminal'\') AH=5B: Create New File (compare INT 21,3C) CX=attrib DS:DX->ASCIIIZ filespec AH=5C: Lock/Unlock File AL=00: Lock =01: Unlock BX : Handle CX,DX: Offset (4 byte) SI,DI: Length (4 byte) AX=5E00: Get Machine Name DS:DX->15 byte area AX=5E02: Set Printer Setup BX=index CX=length of initialize string DS:DI->initialize string to be sent ahead of all files sent to printer AX=5E03: Get Printer Setup BX=index DS:SI->buffer AX=5F02: Get Assign-List Entry BX=index (no. of current redirection) =0: initial index, to be incremented until AX returns error code 18d DS:SI->16 byte buffer ES:DI->128 byte buffer AX=5F03: Make Network Connection BL=3: to printer =4: to disk drive CX=parameter (=00) DS:SI->ASCIIIZ device name (PRN, LPT1, ...) ES:DI->ASCIIIZ network path AX=5F04: Delete Network Connection DS:SI->ASCIIIZ device name and path AH=62: Get PSP address	CF=1: AX=error code CF=0: DS:DX->ASCIIIZ filespec with filename supplied by DOS IF file exists: error (CF=1) CF=1: AX=error code CF=0: AX=handle file opened CF=1: AX=error code CF=1: AX=error code CF=0: CH=0: Name not defined CH<>0: CL='Name of NETBIOS' (?) DS:DX->15 byte ASCIIIZ Machine name CF=1: AX=error code CF=1: AX=error code CF=0: CX=length of initialize string DS:DI->initialize string CF=1: AX=error code =18d: end of redirection list CF=0: BH=status flag: bit 0=0: device unavailable =1: device available BL=device type =3: printer =4: file CX=parameter value DS:SI->ASCIIIZ local device name (PRN, LPT1, ...) ES:DI->ASCIIIZ network name CF=1: AX=error code CF=1: AX=error code BX=segment address

DOS 3.3:

AH=65: Get Extended National Info AL=info type BX=code page CX=no. of data to be returned DX=national ID ES:DI->buffer AX=6601: Get Code Page AX=6602: Change Code Page like DOS command :>CHCP needs installed NLSFUNC	ES:DI->tabel with extended national info CF=1: AX=error code CF=0: BX=active code page DX=system code page
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AH=67: **Set Maximal Handle Count**
BX=maximal open files handle count
 <20: =20
AH=68: **Commit File**
 Empty file buffer
BX=handle

CF=1: AX=error code